

Adopted Levels, Gammas

$Q(\beta^-)=167.322\ 26$; $S(n)=6985.84\ 4$; $S(p)=11586.5\ 8$; $Q(\alpha)=-8322.09\ 6$ [2021Wa16](#)

$S(2n)=18402.99\ 4$, $S(2p)=22909.8\ 7$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Cl}(n,X)$ reaction at Physical Institute of Aarhus University ([1936An01](#)).

^{35}S decay measurements:

[2014Si21](#), [2000Ho13](#), [1999Pa18](#), [1999Oh09](#).

^{35}S cross-section measurements:

$^{35}\text{Cl}(n,p)^{35}\text{S}$: [2024Ha18](#), [2023De34](#), [2021WaZY](#), [2020Ku17](#), [2019Ba16](#), [2000Ka30](#), [1999Gl09](#), [1999Gl02](#);

Projectile fragmentation $^9\text{Be}(^{40}\text{Ar},X)$ and $^{181}\text{Ta}(^{40}\text{Ar},X)$: [2012Zh06](#), [2007No13](#);

Spallation $^{136}\text{Xe}(p,X)$: [2007Na31](#);

$^{34}\text{S}(n,\gamma)$ E=thermal: [2023KoZS](#), [2008FiZZ](#).

 ^{35}S LevelsCross Reference (XREF) Flags

A	$^{35}\text{P}\ \beta^-$ decay (47.3 s)	F	$^{34}\text{S}(n,\gamma)$ E=thermal	K	$^{37}\text{Cl}(p,^3\text{He})$
B	$^2\text{H}(^{34}\text{S},p\gamma)$	G	$^{34}\text{S}(n,n),(n,\gamma)$:resonances	L	$^{37}\text{Cl}(d,\alpha\gamma)$
C	$^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$	H	$^{34}\text{S}(\text{pol } d,p)$	M	$^{160}\text{Gd}(^{34}\text{S},X\gamma),(^{37}\text{Cl},X\gamma)$
D	$^{24}\text{Mg}(^{14}\text{N},3p\gamma)$	I	$^{34}\text{S}(d,p)$	N	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$
E	$^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$	J	$^{34}\text{S}(d,p\gamma)$		

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2}</u>	<u>XREF</u>	<u>Comments</u>
0.0	3/2 ⁺	87.61 d 24	ABCDEF HIJKLMN	$\% \beta^- = 100$ $\mu = (+)1.00\ 4$ (1954Bu05 , 2019StZV) $Q = 0.0483\ 3$ (2018Py01 , 2021StZZ) J^π : L=2 from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L-1/2 transfer from analyzing power. J=3/2 also from microwave spectroscopy (1949Co12 , 1951We11). $T_{1/2}$: unweighted average of 88 d 3 (1941Ka01), 87.1 d 12 (1943He01), 86.35 d 17 (1959Co56), 87.16 d 10 (1958Se49), 88.8 d 10 (1959Ca12), 89.0 d 5 (1961Wy01), 87.1 d 9 (1961Oz01), 87.9 d 3 (1965Fl02), 87.39 d 10 (1968Wo06), 87.5 d 4 (1969La34), and 87.38 d 3 (1999Pa18). μ : +1.00 4 or -1.07 4 from 1954Bu05 using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes. Q : from 2018Py01 using the microwave absorption method. -0.483 3 is given in 2021StZZ . Other: 0.045 from 1954Bi40 using the microwave absorption method.
1572.370 10	1/2 ⁺	2.29 ps 14	ABCD F HIJKL N	J^π : L=0 from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$. $T_{1/2}$: lifetime $\tau = 3.3$ ps 2 in $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$ from 2022Gr07 with DRDM. Others: 3.3 ps 5 in ($^{34}\text{S},p\gamma$) from 1973Wa10 with DSAM, >1.8 ps in (d,p γ) from 1970Bu18 with DSAM, and >1.6 ps in (d,p γ) from 1972Fr11 with DSAM.
1991.27 4	7/2 ⁻	1.03 ns 5	CDEF HIJKLMN	J^π : L(pol d,p)=3 from 0^+ and L+1/2 transfer from analyzing power. $T_{1/2}$: lifetime $\tau = 1.48$ ns 7: weighted average of 1.7 ns 3 in (d,p γ) from 2024Co04 with p γ -delayed coin and 1.47 ns 7 in (d,p γ) from 1971Pr11 with p γ -delayed coin. Other: >4.5 ps in (d,p γ) from 1970Bu18 with DSAM.
2347.779 10	3/2 ⁻	0.78 ps 12	BCD F HIJ L N	J^π : L=1 from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2 transfer from analyzing power. $T_{1/2}$: lifetime $\tau = 1.12$ ps 18: weighted average of 1.3 ps 2 in ($^{34}\text{S},p\gamma$) from 1973Wa10 with DSAM, 0.8 ps 2 in (d,p γ) from 1970Bu18 with DSAM, and 1.22 ps 18 in (d,p γ) from 1972Fr11 with DSAM.
2717.07 9	5/2 ⁺	69 fs 24	CD F HIJKL N	J^π : L=2 from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
				transfer from analyzing power.
2938.62 5	(3/2) ⁺		A C F HIJKL	T _{1/2} : lifetime $\tau=100$ fs 35 in (d,p γ) from 1972Fr11 with DSAM.
3423.2 20	5/2 ⁺	<69 fs	C HIJKL N	J ^π : L=2 from 0 ⁺ in (d,p) and (pol d,p); possibly allowed β^- feeding from 1/2 ⁺ parent with log ft=5.0. Other: L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ is inconsistent.
3558.079 28	(3/2 ⁻ ,5/2 ⁺)		F HIJ L N	J ^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2 transfer from analyzing power.
3594.62 24	7/2 ⁽⁺⁾		CDE IJKL N	T _{1/2} : lifetime $\tau<100$ fs in (d,p γ) from 1972Fr11 with DSAM.
				J ^π : 1986 γ to 1572.370, 1/2 ⁺ and 1566.7 γ to 1991.27, 7/2 ⁻ .
				XREF: C(3602)I(3596)K(3598)
				J ^π : 3594.5 γ Q, $\Delta J=2$ to g.s., 3/2 ⁺ ; L(p, ^3He)=2+4 from 3/2 ⁺ suggests $\pi=+$. Other: L=2 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ is inconsistent.
3675? 10	(1/2 ⁻ ,3/2 ⁻)		I	J ^π : L=(1) from 0 ⁺ in (d,p).
3801.952 16	3/2 ⁻	25 fs 18	F HIJKL	XREF: k(3811)
				J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
				T _{1/2} : lifetime $\tau=36$ fs 26 in (d,p γ) from 1972Fr11 with DSAM.
3816.01 21	(9/2) ⁻	0.28 ps 3	CDE H JkL N	XREF: k(3811)
				J ^π : 1824.7 γ M1+E2 to 1991.27, 7/2 ⁻ ; 2061.6 γ D, $\Delta J=1$ from 5877.70 (11/2 ⁺).
3886.28 33	(5/2 ⁺)		CD IJ L	T _{1/2} : in ($^{14}\text{N},3p\gamma$) from 2014Ay01 with DSAM.
				XREF: C(3895)
				J ^π : 1894.9 γ D, $\Delta J=1$ to 1991.27, 7/2 ⁻ ; 3886.0 γ to g.s., 3/2 ⁺ ; possible 2313.83 γ to 1572.370, 1/2 ⁺ level. Other: L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ is inconsistent; it might be a different level.
4023.24 21	(11/2) ⁻	0.32 ps 3	DE LMN	J ^π : 2032.0 γ E2, $\Delta J=2$ to 1991.27, 7/2 ⁻ ; 207.0 γ to 3816.01, (9/2) ⁻ .
4027.2 14	3/2 ⁺ ,5/2 ⁺		C IJKL	T _{1/2} : in ($^{14}\text{N},3p\gamma$) from 2014Ay01 with DSAM.
				XREF: I(4025)
4105.6 8	(3/2 ⁻ ,5/2 ⁺)	<55 fs	F IJKL	J ^π : L=2 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$.
4180 3	(1/2,3/2,5/2 ⁺)		kL	J ^π : 2533.1 γ to 1572.370, 1/2 ⁺ and 2114.3 γ to 1991.27, 7/2 ⁻ .
				T _{1/2} : lifetime $\tau<80$ fs in (d,p γ) from 1972Fr11 with DSAM.
4189.238 33	1/2 ⁻	<35 fs	F HIJKL	XREF: k(4186)
				J ^π : 2608 γ to 1572.370, 1/2 ⁺ .
				XREF: k(4186)L(4187)
				J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power.
4302.6 7			I KL	T _{1/2} : lifetime $\tau<50$ fs in (d,p γ) from 1972Fr11 with DSAM.
4477.63 7	(5/2 ⁺)	<62 fs	F IJKL	XREF: K(4290)
				XREF: i(4481)K(4489)
				J ^π : 2905.1 γ to 1572.370, 1/2 ⁺ and 2486.27 γ to 1991.27, 7/2 ⁻ .
				L=2 from 3/2 ⁺ in (p, ^3He). L=2 component of L=2,3 doublet from 0 ⁺ in (d,p).
4482	7/2 ⁻		Hi	T _{1/2} : lifetime $\tau<90$ fs in (d,p γ) from 1972Fr11 with DSAM.
				XREF: i(4481)
				J ^π : L=3 from 0 ⁺ in (pol d,p) and L+1/2 transfer from analyzing power. L=3 component of L=2,3 doublet from 0 ⁺ in (d,p).
4576 8	(⁺)		HI K	E(level): weighted average of 4575 8 from (d,p) and 4577 10 from (p, ^3He).

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
4617 10			K	J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He). L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
4822.79 24	(9/2 ⁺)		DE	J ^π : 1228.1γ D, ΔJ=1 to 3594.62, 7/2 ⁽⁺⁾ ; 1055.1γ D, ΔJ=1 from 5877.70, (11/2 ⁺); 9/2 ⁺ from shell-model calculations (2021Go09).
4839 8	(⁺)		HI K	E(level): weighted average of 4837 8 from (d,p) and 4843 10 from (p, ³ He). Other: 4837 from (pol d,p). J ^π : L=2 from 3/2 ⁺ in (p, ³ He) suggests π=+. L=2 gives 1/2 ⁺ to 9/2 ⁺ .
4899.82 27	(9/2 ⁺)		DE	J ^π : 2908.3γ D, ΔJ=1 to 1991.27, 7/2 ⁻ ; 977.9γ D+Q, ΔJ=1 from 5877.70, (11/2 ⁺); 9/2 ⁺ from shell-model calculations (2021Go09).
4903.371 17	1/2 ⁻		F HI	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power.
4963.081 25	3/2 ⁻		F HI K	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
4990 10	(⁺)		K	J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He). L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
5010.05 29	(11/2) ⁻	0.45 ps 8	DE	J ^π : 3018.5γ E2, ΔJ=2 to 1991.27, 7/2 ⁻ ; 986.8γ to 4023.24, (11/2) ⁻ . T _{1/2} : in (¹⁴ N,3pγ) from 2014Ay01 with DSAM.
5058 8	7/2 ⁻		HI	E(level): from (d,p). J ^π : L=3 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
5127 10	(⁺)		HI K	E(level): from (p, ³ He). Other: 5126 11 from (d,p). J ^π : L=2 from 3/2 ⁺ in (p, ³ He) suggests π=+. L=2 gives (1/2 ⁺ to 9/2 ⁺).
5343 8	(⁻)		I K	E(level): weighted average of 5342 8 from (d,p) and 5345 10 from (p, ³ He). J ^π : L=3 from 3/2 ⁺ in (p, ³ He) suggests π=-. L=3 gives (1/2 ⁻ to 11/2 ⁻).
5412.28 25	(9/2 ⁺)		DE	J ^π : 1389.0γ D, ΔJ=1, to 4023.24, (11/2) ⁻ ; 1525.7γ to 3886.28, (5/2 ⁺).
5550 10			I K	XREF: I(5542?)
5752.5 8	5/2 ⁻ , 7/2 ⁻		F I	J ^π : L=(3) from 3/2 ⁺ in (p, ³ He). L=3 gives (1/2 ⁻ to 11/2 ⁻). XREF: I(5740)
5771 10	(⁺)		K	J ^π : L=3 from 0 ⁺ in (d,p). J ^π : L=2 from 3/2 ⁺ in (p, ³ He) suggests π=+. L=2 gives (1/2 ⁺ to 9/2 ⁺).
5841.482 31			F	
5877.70 22	(11/2 ⁺)		DE	J ^π : 465.4γ D, ΔJ=1 to 5412.28, (9/2 ⁺); 867.6γ D, ΔJ=0 to 5010.05, (11/2) ⁻ ; strong 2283.0γ to 3594.62, 7/2 ⁽⁺⁾ .
5890 20	3/2 ⁺ , 5/2 ⁺		I K	XREF: K(5915?) J ^π : L=2 from 0 ⁺ in (d,p).
5980 10			I	
6018.8 6			F	
6078.475 33	1/2 ⁻ , 3/2 ⁻		F I	J ^π : L=1 from 0 ⁺ in (d,p).
6129 10	(⁺)		K	J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He) suggests π=+. L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
6293.93 6			F I	XREF: I(6292)
6299.4 4	(11/2)		D	E(level): other: 6292 8 from (d,p).
6334 8			I	J ^π : 887.0γ, ΔJ=1 to 5412, (9/2 ⁺); 2275.9γ to 4023.24, (11/2) ⁻ .
6344 8	(⁺)		I K	XREF: K(6347)
6352.2 5	(13/2) ⁻	50 fs 10	D	J ^π : L=(2) from 3/2 ⁺ in (p, ³ He) suggests π=(+). J ^π : 2535.8γ E2, ΔJ=2 to 3816.01, (9/2) ⁻ ; 2329.5γ (M1), ΔJ=1 to

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
				4023.24, (11/2) ⁻ . T _{1/2} : in ($^{14}\text{N}, 3\text{py}$) from 2014Ay01 with DSAM.
6354.89 23			F	
6419.9 11			F	
6446 8			I	
6496 8			I	
6537.7 14			I	
6545.1 13			I	
6584 10			I	
6629.43 9			F	
6635.2 13			I	
6654 10	(⁻)		K	J ^π : L=(3) from 3/2 ⁺ in (p, ^3He) suggests $\pi=(-)$.
6684 9	(⁺)		I K	E(level): weighted average of 6677 8 from (d,p) and 6696 10 from (p, ^3He). J ^π : L=2 from 3/2 ⁺ in (p, ^3He) suggests $\pi=+$. L=2 gives (1/2 ⁺ to 9/2 ⁺).
6761.0 12			F	
6891.3 14	(⁻)		I	J ^π : L=(1,3) from 0 ⁺ in (d,p).
(6986.096 25)	1/2 ⁺		F	J ^π : s-wave resonance capture state.
7018.89 4	3/2 ⁺ , 5/2 ⁺		G I	E(level): from (n,n). Other: 7021 10 from (d,p).
7072.66 5			G	
7074.33 6			G	
7097.72 30	3/2 ⁺ , 5/2 ⁺		G	
7099.85 20	3/2 ⁺ , 5/2 ⁺		G	
7100.73 4	3/2 ⁻		G	
7143.27 11			G I K	XREF: I(7150)K(7151)
7179.74 32	(15/2 ⁺)	3.1 ps 12	DE	J ^π : 1302.0γ E2, ΔJ=2 to 5877.70, (11/2 ⁺); 827.8γ to 6352.2, (13/2) ⁻ . T _{1/2} : in ($^{14}\text{N}, 3\text{py}$) from 2014Ay01 with DSAM. XREF: I(7205)
7210.42 5			G I	
7218.0 4			G	
7234.0 5			G	
7239.8 6			G	
7253.2 6			G	
7275.93 5	1/2 ⁺		G	
7289.8 8			G	
7294.19 6	3/2 ⁻		G	
7331.00 5	1/2 ⁺		G	
7338.18 20			G	
7343.91 20			G	
7347.2 5			G	
7347.9 5			G	
7353.9 5			G	
7357.6 5			G	
7370.56 6	1/2 ⁻		G K	XREF: K(7375)
7396.0 6			G	
7404.6 6			G	
7408.68 11	3/2 ⁻		G	
7411.55 6			G	
7416.50 4			G	
7429.1 6			G	
7433.56 5			G	
7436.4 4			G	
7442.14 5	1/2 ⁺		G	
7462.2 6			G	
7481.15 7	1/2 ⁻		G I	E(level): weighted average of 7481.15 6 from (n,n) and 7482.7 13

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
				from (d,p).
7494.5 6			G	
7542.3 6			G i	XREF: i(7570)
7604.0 6			G i	XREF: i(7570)
7609.26 5	3/2 ⁻		G	
7648.9 6			G i	XREF: i(7640)
7655.7 6			G i	XREF: i(7640)
7663.91 11			G	
7678.3 7			G	
7731.1 7			G K	XREF: K(7712?)
7749.7 4	(3/2)		G	
7761.46 7	3/2 ⁻		G k	XREF: k(7770)
7776.17 11	1/2 ⁻		G k	XREF: k(7770)
7797.99 9	1/2 ⁺		G	
7812.24 7	3/2 ⁺		G	
7853.25 7	3/2 ⁻		G	
7861.8 12			G	
7880.8 12			G	
7889.0 12			G	
7894.63 7	3/2 ⁻		G	
7899.70 30			G	
7934.1 15			G	
7939.5 15			G	
7954.96 11	3/2 ⁻		G	
7974.19 11	3/2 ⁻		G	
8019.5 2	3/2 ⁺ , 5/2 ⁺		G	
8024.0 7	(17/2 ⁺)	0.15 ps 4	DE	J ^π : 844.2γ (M1), ΔJ=1 to 7179.74, (15/2 ⁺). 17/2 ⁺ from shell-model calculations (2021Go09). T _{1/2} : in (¹⁴ N,3pγ) from 2014Ay01 with DSAM.
8041.20 20	3/2 ⁺ , 5/2 ⁺		G	
8076.46 20	3/2 ⁺ , 5/2 ⁺		G	
8077.91 30	3/2 ⁺ , 5/2 ⁺		G	
8093.0 5	1/2 ⁻		G K	XREF: K(8103) E(level): from (n,n). Other: 8103 10 from (p, ³ He).
8143.75 20			G	
8160 10			K	
8187.17 30	1/2 ⁻		G	
8211.25 30			G	
8224.2 4	1/2 ⁻		G	
8228.3 4	3/2 ⁺ , 5/2 ⁺		G	
8243.98 30	3/2 ⁺ , 5/2 ⁺		G	
8256.31 20	3/2 ⁻		G	
8262.4 4	5/2 ⁺		G	
8273.3 4	(1/2 ⁻)		G	
8298.27 30	3/2 ⁺ , 5/2 ⁺		G	
8333.8 4	3/2 ⁻		G	
8335.9 4	3/2 ⁺ , 5/2 ⁺		G	
8391.3 4	3/2 ⁻		G	
8393.64 30	3/2 ⁺ , 5/2 ⁺		G	
8406.55 30	3/2 ⁺ , 5/2 ⁺		G k	XREF: k(8430)
8418.8 4	3/2 ⁺ , 5/2 ⁺		G k	XREF: k(8430)
8755.9 11	(17/2)	<1 ps	E	J ^π : 1576.3γ D, ΔJ=1 to 7179.74, (15/2 ⁺). T _{1/2} : in (¹⁸ O,2αnγ) from 2021Go09 with residual Doppler energy shifts observed for the 732γ and 1576γ peaks.
9155 10			K	J ^π : L=2 from 3/2 ⁺ in (p, ³ He). L=2 gives (1/2 ⁺ to 9/2 ⁺).

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Adopted Levels, Gammas (continued)

³⁵S Levels (continued)

E(level) [†]	J ^π [‡]	XREF	Comments
10048.6 9	(19/2 ⁺)	E	J ^π : 2868.7γ Q, ΔJ=2 to (15/2 ⁺).
12469.3 14	(21/2)	E	J ^π : 2420.6γ (D), ΔJ=(1) to (19/2).

[†] From a least-squares fit to γ-ray energies with uncertainties for levels connected with γ transitions; from transfer reactions or ³⁴S(n,n),(n,γ):resonances for other levels, unless otherwise noted.

[‡] For high-spin states, it is assumed that spin increases as the excitation energy increases. For resonances (>7 MeV), J^π from ³⁴S(n,n),(n,γ):resonances.

Adopted Levels, Gammas (continued)

<u>E_i(level)</u>	<u>J_i^π</u>	<u>γ(³⁵S)</u>		<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>δ[‡]</u>	<u>α[#]</u>	<u>Comments</u>
		<u>E_γ[†]</u>	<u>I_γ[†]</u>						
1572.370	1/2 ⁺	1572.327 12	100	0.0	3/2 ⁺	(M1)		0.0001007 14	B(M1)(W.u.)=0.00247 +16-14 E _γ : weighted average of 1572.252 29 from ³⁵ P β ⁻ decay, 1576 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 1572.4 3 from (¹⁴ N,3pγ), 1572.333 8 from (n,γ) E=thermal, 1572.24 17 from (d,pγ), and 1572 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
1991.27	7/2 ⁻	1991.26 5	100	0.0	3/2 ⁺	M2+E3	+0.17 6	0.0001476 22	Mult.: D from (¹⁴ N,3pγ); Δπ=no from level scheme. B(M2)(W.u.)=0.087 5; B(E3)(W.u.)=3.7 +28-22 E _γ : weighted average of 1990 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 1991.3 2 from (¹⁴ N,3pγ), 1990.5 4 from (¹⁸ O,2αnγ), 1991.27 5 from (n,γ) E=thermal, and 1991 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). Mult.,δ: from (d,pγ). D+Q or Q+O from γ(θ) in (d,pγ); E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2 ⁻ to 3/2 ⁺ .
2347.779	3/2 ⁻	356.66@ 9 775.398 6	<0.084 35.3 27	1991.27 1572.370	7/2 ⁻ 1/2 ⁺	[E2] (E1)		1.15×10 ⁻³ 2 3.54×10 ⁻⁵ 5	E _γ ,I _γ : from (n,γ) E=thermal. B(E1)(W.u.)=4.5×10 ⁻⁴ +9-7 E _γ : from (n,γ) E=thermal. Others: 778 2 from ⁹ Be(³⁶ S, ³⁵ Sγ) and 775.4 4 from (¹⁴ N,3pγ). I _γ : weighted average of 37 5 from (¹⁴ N,3pγ), 35 4 from (n,γ) E=thermal, 37.0 28 from (d,pγ), and 33.3 27 from (d,αγ). Mult.: D(+Q) from γ(θ) in (d,pγ); Δπ=yes from level scheme; M2 ruled out by RUL.
		2347.700 19	100.0 27	0.0	3/2 ⁺	(E1)		0.000880 12	B(E1)(W.u.)=4.6×10 ⁻⁵ +9-6 E _γ : weighted average of 2352 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 2347.8 3 from (¹⁴ N,3pγ), 2347.700 15 from (n,γ) E=thermal, and 2347 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from (d,αγ). Others: 100 8 from (¹⁴ N,3pγ), 100 8 from (n,γ) E=thermal, and 100.0 28 from (d,pγ). Mult.: D(+Q) from γ(θ) in (d,pγ); Δπ=yes from level scheme; M2 component is less likely based on RUL.
2717.07	5/2 ⁺	368.7 5	14 8	2347.779	3/2 ⁻	[E1]		0.0002294 33	B(E1)(W.u.)=0.022 +19-10 E _γ : weighted average of 368.5 4 from (n,γ) E=thermal and 370 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : unweighted average of 6.7 23 from (n,γ) E=thermal and 22 4 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
		725.79	<5	1991.27	7/2 ⁻	[E1]		4.10×10 ⁻⁵ 6	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	Comments
2717.07	5/2 ⁺	1144.68 2717.01 16	<5 100 10	1572.370 0.0	1/2 ⁺ 3/2 ⁺	[E2] M1+E2	-0.39 5	3.79×10 ⁻⁵ 5 0.000568 9	B(M1)(W.u.)=0.012 +7-3; B(E2)(W.u.)=0.9 +6-3 E _γ : weighted average of 2720 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 2717.0 4 from (¹⁴ N,3pγ), 2716.99 16 from (n,γ) E=thermal, and 2717 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). Other: 100 17 from (n,γ) E=thermal. Mult.,δ: D+Q from γ(θ) in (d,pγ) with δ=-0.39 5 or -15.0 34 for J _i =5/2 (1975VaYG); -15.0 34 is unlikely based on RUL; E1+M2 is ruled out by RUL.
2938.62	(3/2) ⁺	590.84 947.34 1366.22 2938.57 18	<10 <20 <15 100	2347.779 1991.27 1572.370 0.0	3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺				E _γ : weighted average of 2938.29 40 from ³⁵ P β ⁻ decay, 2943 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), and 2938.58 11 from (n,γ) E=thermal.
3423.2	5/2 ⁺	484.6 706.1 1075.4 1431.9 1850.8 3423 2	<4 <6 <7 <30 <7 100	2938.62 2717.07 2347.779 1991.27 1572.370 0.0	(3/2) ⁺ 5/2 ⁺ 3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺	[M1,E2] [M1,E2] [E1] [E1] [E2] [M1,E2]		2.7×10 ⁻⁴ 13 9.5×10 ⁻⁵ 28 1.830×10 ⁻⁵ 26 0.0002257 32 0.000255 4 0.00090 7	E _γ : weighted average of 3425 2 from ⁹ Be(³⁶ S, ³⁵ Sγ) and 3421 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from (n,γ) E=thermal. E _γ : from (n,γ) E=thermal. Other: 1211 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : weighted average of 54 6 from (n,γ) E=thermal and 54 6 from (d,αγ). Other: 18 7 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ : from (n,γ) E=thermal. Other: 1567 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from (d,αγ). Others: 100 19 from (n,γ) E=thermal and 100 10 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from (n,γ) E=thermal.
3558.079	(3/2 ⁻ ,5/2 ⁺)	619.23 19 1210.28 4	8.9 17 54 6	2938.62 2347.779	(3/2) ⁺ 3/2 ⁻				
		1566.7 3	100 6	1991.27	7/2 ⁻				
		1986 & 2 3558.1 5	39 6 18 4	1572.370 0.0	1/2 ⁺ 3/2 ⁺				

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{S})$ (continued)</u>									
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>$\delta^\ddagger$</u>	<u>$\alpha^\#$</u>	<u>Comments</u>
3594.62	7/2 ⁽⁺⁾	655.99	<10	2938.62	(3/2) ⁺				E_γ, I_γ : from ($^{14}\text{N}, 3p\gamma$). Other: $I_\gamma < 11$ from (d, p γ). E_γ : weighted average of 3594.7 7 from ($^{14}\text{N}, 3p\gamma$), 3594.4 11 from ($^{18}\text{O}, 2\alpha n\gamma$), and 3593 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : from ($^{14}\text{N}, 3p\gamma$). Other: 3602 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult.: from $\gamma\gamma(\theta)$ (ADO) in ($^{14}\text{N}, 3p\gamma$). B(E1)(W.u.) = $2.4 \times 10^{-4} + 29 - 12$ E_γ, I_γ : from (n, γ) E=thermal. Other: $I_\gamma < 8.9$ from (d, p γ). B(E1)(W.u.) = $2.0 \times 10^{-4} + 25 - 11$ E_γ : from (n, γ) E=thermal. I_γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from (d, p γ). E_γ : from (n, γ) E=thermal. I_γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from (d, p γ). B(M1)(W.u.) = $0.024 + 29 - 12$ if M1, B(E2)(W.u.) = $4 \times 10^1 + 5 - 2$ if E2. B(E1)(W.u.) = $9 \times 10^{-4} + 11 - 4$ E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from (d, p γ), and 61 5 from (d, $\alpha\gamma$). B(E1)(W.u.) = $2.2 \times 10^{-4} + 25 - 10$ E_γ : from (n, γ) E=thermal. I_γ : from (d, $\alpha\gamma$). Other: 100 9 from (d, p γ), 100 9 from (n, γ) E=thermal. B(M1)(W.u.) = $0.0099 + 14 - 12$; B(E2)(W.u.) = $3.4 + 10 - 9$ E_γ : weighted average of 1824.6 2 from ($^{14}\text{N}, 3p\gamma$), 1824.4 6 from ($^{18}\text{O}, 2\alpha n\gamma$), and 1827 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 1830 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult., δ : D+Q and δ from $\gamma(\theta)$ and
		877.6 5	9.3 23	2717.07	5/2 ⁺				
		1246.82	<15	2347.779	3/2 ⁻				
		1603.31	<17	1991.27	7/2 ⁻				
3801.952	3/2 ⁻	3594.5 7	100 9	0.0	3/2 ⁺	Q			E_γ, I_γ : from (n, γ) E=thermal. Other: $I_\gamma < 8.9$ from (d, p γ). B(E1)(W.u.) = $2.0 \times 10^{-4} + 25 - 11$ E_γ : from (n, γ) E=thermal. I_γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from (d, p γ). E_γ : from (n, γ) E=thermal. I_γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from (d, p γ). B(M1)(W.u.) = $0.024 + 29 - 12$ if M1, B(E2)(W.u.) = $4 \times 10^1 + 5 - 2$ if E2. B(E1)(W.u.) = $9 \times 10^{-4} + 11 - 4$ E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from (d, p γ), and 61 5 from (d, $\alpha\gamma$). B(E1)(W.u.) = $2.2 \times 10^{-4} + 25 - 10$ E_γ : from (n, γ) E=thermal. I_γ : from (d, $\alpha\gamma$). Other: 100 9 from (d, p γ), 100 9 from (n, γ) E=thermal. B(M1)(W.u.) = $0.0099 + 14 - 12$; B(E2)(W.u.) = $3.4 + 10 - 9$ E_γ : weighted average of 1824.6 2 from ($^{14}\text{N}, 3p\gamma$), 1824.4 6 from ($^{18}\text{O}, 2\alpha n\gamma$), and 1827 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 1830 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult., δ : D+Q and δ from $\gamma(\theta)$ and
		863.28 28	1.3 4	2938.62	(3/2) ⁺	[E1]		2.82×10^{-5} 4	
		1084.79 15	2.2 7	2717.07	5/2 ⁺	[E1]		1.801×10^{-5} 25	
		1454.09 4	18 5	2347.779	3/2 ⁻	[M1, E2]		8.0×10^{-5} 12	
		1810.63	<8.9	1991.27	7/2 ⁻	[E2]		0.0002364 33	
		2229.510 16	88 15	1572.370	1/2 ⁺	[E1]		0.000804 11	
		3801.73 4	100 5	0.0	3/2 ⁺	[E1]		1.60×10^{-3} 2	
3816.01	(9/2) ⁻	1824.7 3	100	1991.27	7/2 ⁻	M1+E2	+0.55 9	0.000200 4	E_γ, I_γ : from (n, γ) E=thermal. Other: $I_\gamma < 8.9$ from (d, p γ). B(E1)(W.u.) = $2.0 \times 10^{-4} + 25 - 11$ E_γ : from (n, γ) E=thermal. I_γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from (d, p γ). E_γ : from (n, γ) E=thermal. I_γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from (d, p γ). B(M1)(W.u.) = $0.024 + 29 - 12$ if M1, B(E2)(W.u.) = $4 \times 10^1 + 5 - 2$ if E2. B(E1)(W.u.) = $9 \times 10^{-4} + 11 - 4$ E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from (d, p γ), and 61 5 from (d, $\alpha\gamma$). B(E1)(W.u.) = $2.2 \times 10^{-4} + 25 - 10$ E_γ : from (n, γ) E=thermal. I_γ : from (d, $\alpha\gamma$). Other: 100 9 from (d, p γ), 100 9 from (n, γ) E=thermal. B(M1)(W.u.) = $0.0099 + 14 - 12$; B(E2)(W.u.) = $3.4 + 10 - 9$ E_γ : weighted average of 1824.6 2 from ($^{14}\text{N}, 3p\gamma$), 1824.4 6 from ($^{18}\text{O}, 2\alpha n\gamma$), and 1827 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 1830 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult., δ : D+Q and δ from $\gamma(\theta)$ and

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$\gamma(^{35}\text{S})$ (continued)							Comments
		$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	
3886.28	(5/2 ⁺)	463.1	<30	3423.2	5/2 ⁺				$\gamma\gamma(\theta)(\text{ADO})$ in (¹⁴ N,3p γ); E1+M2 ruled out by RUL.
		947.65	<12	2938.62	(3/2) ⁺				
		1169.19	<15	2717.07	5/2 ⁺				
		1538.4 8	57 33	2347.779	3/2 ⁻				
4023.24	(11/2) ⁻	1894.9 4	100 11	1991.27	7/2 ⁻	D			E γ : from (¹⁴ N,3p γ). Other: 1542 2 from ⁹ Be(³⁶ S, ³⁵ S γ). I γ : unweighted average of 24 10 from (¹⁴ N,3p γ) and 89 9 from (d, $\alpha\gamma$). Other: <18 from (d,p γ). E γ ,Mult.: from (¹⁴ N,3p γ). I γ : from (d, $\alpha\gamma$). Other: 100 14 from (¹⁴ N,3p γ) and 100 from (d,p γ). I γ : from (d, $\alpha\gamma$). Other: <16 from (d,p γ). B(M1)(W.u.)=0.046 +25-22 if M1. δ : calculated by evaluators from the measured half-life of the level and RUL=100 for B(E2)(W.u.). B(E2)(W.u.)=7.5 +8-7 E γ : weighted average of 2031.8 3 from (¹⁴ N,3p γ) and 2032.4 4 from (¹⁸ O,2 α n γ). Other: 2032 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ S γ). Mult.: Q from $\gamma\gamma(\theta)(\text{ADO})$ in (¹⁴ N,3p γ) and M2 ruled out by RUL.
		2313.83	<16	1572.370	1/2 ⁺				
		3886.05	33 7	0.0	3/2 ⁺	[M1+E2]	<0.25	0.005 4	
		207.0 5	0.59 30	3816.01	(9/2) ⁻				
4027.2	3/2 ⁺ ,5/2 ⁺	2032.0 3	100 6	1991.27	7/2 ⁻	E2		0.000341 5	I γ : weighted average of 86 22 from (d,p γ) and 97 12 from (d, $\alpha\gamma$). E γ : from ⁹ Be(³⁶ S, ³⁵ S γ). I γ : from (d, $\alpha\gamma$). Other: 92 33 from (d,p γ). E γ : from ⁹ Be(³⁶ S, ³⁵ S γ). I γ : from (d, $\alpha\gamma$). Other: 100 33 from (d,p γ).
		469.1	<14	3558.079	(3/2 ⁻ ,5/2 ⁺)				
		604.0	<28	3423.2	5/2 ⁺				
		1088.6	95 12	2938.62	(3/2) ⁺				
4105.6	(3/2 ⁻ ,5/2 ⁺)	1309 2	<22	2717.07	5/2 ⁺				E γ : from (n, γ) E=thermal.
		1679.4	97 12	2347.779	3/2 ⁻				
		2037 2	<28	1991.27	7/2 ⁻				
		2454.7	100 18	1572.370	1/2 ⁺				
		4027.0	<36	0.0	3/2 ⁺				
		547.5	<4.6	3558.079	(3/2 ⁻ ,5/2 ⁺)				
		682.4	<4.6	3423.2	5/2 ⁺				
		1167.0	15 7	2938.62	(3/2) ⁺				
		1388.5	<12	2717.07	5/2 ⁺				
		1757.8	<5.8	2347.779	3/2 ⁻				
		2114.3	<8.1	1991.27	7/2 ⁻				
		2533.1	<5.8	1572.370	1/2 ⁺				
		4105.3 8	100 7	0.0	3/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
4180	(1/2,3/2,5/2 ⁺)	2608	22 6	1572.370	1/2 ⁺			I_γ : from (d, $\alpha\gamma$).
		4180	100 6	0.0	3/2 ⁺			I_γ : from (d, $\alpha\gamma$).
4189.238	1/2 ⁻	631.32 @ 24	<2.8	3558.079	(3/2 ⁻ ,5/2 ⁺)			E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <7.7 from (d,p γ).
		1250.61 5	8.4 10	2938.62	(3/2 ⁺)	[E1]	0.0001047 15	E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <9.6 from (d,p γ).
		1840.2 12	91 10	2347.779	3/2 ⁻	[M1,E2]	0.000221 29	E_γ : from (n, γ) E=thermal.
								I_γ : weighted average of 87 16 from (n, γ) E=thermal and 92 10 from (d,p γ).
		2616.8 13	9.5 28	1572.370	1/2 ⁺	[E1]	1.04×10 ⁻³ 2	E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <12 from (d,p γ).
		4188.95 5	100 10	0.0	3/2 ⁺	[E1]	1.75×10 ⁻³ 3	E_γ : from (n, γ) E=thermal.
4302.6		1954.8	100 9	2347.779	3/2 ⁻			I_γ : other: 100 11 from (n, γ) E=thermal.
		4302.3	70 9	0.0	3/2 ⁺			I_γ : from (d, $\alpha\gamma$).
4477.63	(5/2 ⁺)	591.35	<8.2	3886.28	(5/2 ⁺)	[M1,E2]	1.5×10 ⁻⁴ 6	I_γ : from (d, $\alpha\gamma$).
		675.67	<8.2	3801.952	3/2 ⁻	[E1]	4.82×10 ⁻⁵ 7	
		883.00	<8.2	3594.62	7/2 ⁽⁺⁾	[M1,E2]	5.4×10 ⁻⁵ 12	
		919.54	<8.2	3558.079	(3/2 ⁻ ,5/2 ⁺)			
		1054.4	<16	3423.2	5/2 ⁺	[M1,E2]	3.6×10 ⁻⁵ 6	
		1538.97	<12	2938.62	(3/2 ⁺)			
		1760.55 11	100 12	2717.07	5/2 ⁺	[M1,E2]	0.000189 25	E_γ : from (n, γ) E=thermal.
								I_γ : others: 100 15 from (n, γ) E=thermal and 100 11 from (d, $\alpha\gamma$).
		2129.78	<13	2347.779	3/2 ⁻	[E1]	0.000738 10	
		2486.27	<16	1991.27	7/2 ⁻	[E1]	0.000962 13	I_γ : other: 42 11 from (d, $\alpha\gamma$).
		2905.1 4	90 14	1572.370	1/2 ⁺	[E2]	0.000749 10	E_γ : from (n, γ) E=thermal.
								I_γ : unweighted average of 96 28 from (n, γ) E=thermal, 64 12 from (d,p γ), and 109 11 from (d, $\alpha\gamma$).
4822.79	(9/2 ⁺)	4477.32	<20	0.0	3/2 ⁺	[M1,E2]	0.00126 8	I_γ : other: 28 6 from (d, $\alpha\gamma$).
		1006.8 2	66 15	3816.01	(9/2 ⁻)	D		E_γ : from (¹⁴ N,3p γ). Other: 1007.2 11 from (¹⁸ O,2 α n γ).
								I_γ : weighted average of 73 9 from (¹⁴ N,3p γ) and 36 18 from (¹⁸ O,2 α n γ).
								Mult.: from $\gamma\gamma(\theta)$ (ADO) in (¹⁴ N,3p γ) and (¹⁸ O,2 α n γ), consistent with $\Delta J=0$.
		1228.1 3	100 9	3594.62	7/2 ⁽⁺⁾	D		E_γ, I_γ : other: 1228.3 6 with $I_\gamma=100$ 18 from (¹⁸ O,2 α n γ).
								Mult.: from (¹⁴ N,3p γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
4822.79	(9/2 ⁺)	2831.4 9	36 9	1991.27	7/2 ⁻			
4899.82	(9/2 ⁺)	1305.3 4	55 18	3594.62	7/2 ⁽⁺⁾			E_γ : from ($^{18}\text{O}, 2\alpha n\gamma$). Other: 1304.8 12 from ($^{14}\text{N}, 3p\gamma$).
		2908.3 5	100 18	1991.27	7/2 ⁻	D		Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ($^{14}\text{N}, 3p\gamma$).
4903.371	1/2 ⁻	1101.92 31	0.43 11	3801.952	3/2 ⁻			
		1964.8 2	4.9 13	2938.62	(3/2) ⁺			
		2555.492 14	41 4	2347.779	3/2 ⁻			
		3330.84 13	100 9	1572.370	1/2 ⁺			
		4902.99 5	53 5	0.0	3/2 ⁺			
4963.081	3/2 ⁻	1161.05 20	0.98 14	3801.952	3/2 ⁻			
		1404.967 24	10.7 11	3558.079	(3/2 ⁻ , 5/2 ⁺)			
		2615.2 2	19.5 18	2347.779	3/2 ⁻			
		2972.0 4	3.2 11	1991.27	7/2 ⁻			
		3390.62 7	100 9	1572.370	1/2 ⁺			
		4963.06 22	47 5	0.0	3/2 ⁺			
5010.05	(11/2) ⁻	986.8 3	100 5	4023.24	(11/2) ⁻	(M1)		B(M1)(W.u.)=0.036 +8-6 E_γ : weighted average of 986.9 3 from ($^{14}\text{N}, 3p\gamma$) and 986.6 3 from ($^{18}\text{O}, 2\alpha n\gamma$). Mult.: D from ($^{18}\text{O}, 2\alpha n\gamma$) and ($^{14}\text{N}, 3p\gamma$); $\Delta\pi$ =no from level scheme.
		3018.5 7	43 5	1991.27	7/2 ⁻	E2	0.000798 11	B(E2)(W.u.)=0.22 +5-4 Mult.: Q from $\gamma\gamma(\theta)(\text{ADO})$ and M2 ruled out by RUL in ($^{14}\text{N}, 3p\gamma$).
5412.28	(9/2 ⁺)	1389.0 4	100 10	4023.24	(11/2) ⁻	D		E_γ : other: E_γ =1389.0 8 from ($^{18}\text{O}, 2\alpha n\gamma$). Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ($^{14}\text{N}, 3p\gamma$).
		1525.7 8	15.4 26	3886.28	(5/2 ⁺)			
		3420.6 9	28.2 26	1991.27	7/2 ⁻			
5752.5	5/2 ⁻ , 7/2 ⁻	5752.0 8	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
5841.482		4268.65 14	100	1572.370	1/2 ⁺			E_γ : weighted average of 465.3 3 from ($^{14}\text{N}, 3p\gamma$) and 465.5 2 from ($^{18}\text{O}, 2\alpha n\gamma$).
5877.70	(11/2 ⁺)	465.4 2	38 6	5412.28	(9/2 ⁺)	D		I_γ : unweighted average of 31.6 26 from ($^{14}\text{N}, 3p\gamma$) and 43.8 32 from ($^{18}\text{O}, 2\alpha n\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha n\gamma$).
		867.6 4	50.0 26	5010.05	(11/2) ⁻	D		E_γ : weighted average of 867.3 3 from ($^{14}\text{N}, 3p\gamma$) and 868.2 4 from ($^{18}\text{O}, 2\alpha n\gamma$). I_γ : other: 62 7 from ($^{18}\text{O}, 2\alpha n\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha n\gamma$).
		977.9 3	43 5	4899.82	(9/2 ⁺)	D+Q		E_γ : weighted average of 977.8 3 from ($^{14}\text{N}, 3p\gamma$) and 978.4 10 from ($^{18}\text{O}, 2\alpha n\gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$
							Comments
5877.70	(11/2 ⁺)	1055.1 6	80 7	4822.79	(9/2 ⁺)	D	I _γ : weighted average of 42.1 26 from (¹⁴ N,3pγ) and 82 22 from (¹⁸ O,2αnγ). Mult.: from (¹⁴ N,3pγ). E _γ : unweighted average of 1054.5 3 from (¹⁴ N,3pγ) and 1055.6 4 from (¹⁸ O,2αnγ). I _γ : weighted average of 79 5 from (¹⁴ N,3pγ) and 122 32 from (¹⁸ O,2αnγ). Mult.: from (¹⁸ O,2αnγ).
		1854.4 4	44.7 26	4023.24	(11/2) ⁻		E _γ ,I _γ : other: 1854.5 42 with I _γ =88 44 from (¹⁸ O,2αnγ).
		2061.6 4	100 5	3816.01	(9/2) ⁻	D	E _γ : weighted average of 2061.5 3 from (¹⁴ N,3pγ) and 2063.3 14 from (¹⁸ O,2αnγ). I _γ : other: 100 20 from (¹⁸ O,2αnγ). Mult.: from (¹⁸ O,2αnγ).
		2283.0 4	56 12	3594.62	7/2 ⁽⁺⁾		E _γ : weighted average of 2282.9 4 from (¹⁴ N,3pγ) and 2283.5 10 from (¹⁸ O,2αnγ). I _γ : unweighted average of 68.4 26 from (¹⁴ N,3pγ) and 44 10 from (¹⁸ O,2αnγ).
6018.8		3886.0 13	7.9 26	1991.27	7/2 ⁻		
6078.475	1/2 ⁻ ,3/2 ⁻	6018.2 6	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
		3139.9 5	22 7	2938.62	(3/2) ⁺		
		6077.87 11	100 12	0.0	3/2 ⁺		
6293.93		6293.2 4	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
6299.4	(11/2)	887.0 7	40 10	5412.28	(9/2 ⁺)	D	
		2275.9 6	100 20	4023.24	(11/2) ⁻		
		2483.6 8	90 20	3816.01	(9/2) ⁻		
6352.2	(13/2) ⁻	2329.5 9	100 20	4023.24	(11/2) ⁻	(M1)	B(M1)(W.u.)=0.023 +6-5 Mult.: D from γγ(θ)(ADO) in (¹⁴ N,3pγ); Δπ=no from level scheme.
		2535.8 11	50 6	3816.01	(9/2) ⁻	E2	B(E2)(W.u.)=5.3 +18-12 Mult.: Q from γγ(θ)(ADO) in (¹⁴ N,3pγ) and M2 ruled out by RUL.
6354.89		6355.0 6	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
6419.9		6419.3 11	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
6629.43		6628.5 6	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
6761.0		6760.3 12	100	0.0	3/2 ⁺		E _γ ,I _γ : from (n,γ) E=thermal.
(6986.096)	1/2 ⁺	356.66 @ 9	<0.073	6629.43			
		631.32 @ 24	<0.123	6354.89			
		692.16 5	0.238 31	6293.93			
		907.608 23	1.00 18	6078.475	1/2 ⁻ ,3/2 ⁻		
		1144.591 20	0.98 11	5841.482			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
(6986.096)	1/2 ⁺	2022.954 9	19.6 18	4963.081	3/2 ⁻			
		2082.83 17	28.4 27	4903.371	1/2 ⁻			
		2508.39 8	0.68 7	4477.63	(5/2 ⁺)			
		2796.74 4	8.8 7	4189.238	1/2 ⁻			
		3183.95 4	10.9 11	3801.952	3/2 ⁻			
		4638.14 14	100 9	2347.779	3/2 ⁻			
		6985.7 10	0.064 14	0.0	3/2 ⁺			
7179.74	(15/2 ⁺)	827.8 6	2.2 8	6352.2	(13/2) ⁻	[E1]	3.08×10 ⁻⁵ 4	B(E1)(W.u.)=7×10 ⁻⁶ +5-3
		880.2 4	5.9 8	6299.4	(11/2)			
		1302.0 3	100 6	5877.70	(11/2 ⁺)	E2	5.56×10 ⁻⁵ 8	B(E2)(W.u.)=7 +4-2 E _γ : weighted average of 1302.2 2 from (¹⁴ N,3pγ) and 1301.6 3 from (¹⁸ O,2αnγ). Mult.: Q from γγ(θ)(ADO) in (¹⁴ N,3pγ) and M2 ruled out by RUL.
8024.0	(17/2 ⁺)	844.2 6	100	7179.74	(15/2 ⁺)	(M1)		B(M1)(W.u.)=0.24 +9-5 E _γ : weighted average of 843.9 8 from (¹⁴ N,3pγ) and 844.3 6 from (¹⁸ O,2αnγ). Mult.: D from γγ(ADO) in (¹⁸ O,2αnγ); Δπ=(no) from level scheme.
8755.9	(17/2)	731.6 15	57 22	8024.0	(17/2 ⁺)			E _γ ,I _γ : from (¹⁸ O,2αnγ).
		1576.3 14	100 22	7179.74	(15/2 ⁺)	D		E _γ ,I _γ : from (¹⁸ O,2αnγ).
10048.6	(19/2 ⁺)	2868.7 8	100	7179.74	(15/2 ⁺)	Q		E _γ ,I _γ : from (¹⁸ O,2αnγ).
12469.3	(21/2)	2420.6 11	100	10048.6	(19/2 ⁺)	(D)		E _γ ,I _γ : from (¹⁸ O,2αnγ).

[†] For γ rays from low-spin levels (J<9/2), values are from (d,pγ) up to 4478 level and from (n,γ) E=thermal above that, for γ rays from high-spin level (J≥9/2), values are from ²⁴Mg(¹⁴N,3pγ), unless otherwise noted. E_γ values without uncertainties are deduced from level-energy differences.

[‡] Mult and δ are primarily from γ(θ) and γγ(ADO) in ³⁴S(d,pγ), ²⁴Mg(¹⁴N,3pγ), and ²⁶Mg(¹⁸O,2αnγ). The magnetic or electric nature is determined based on RUL and measured T_{1/2} where available, unless otherwise noted.

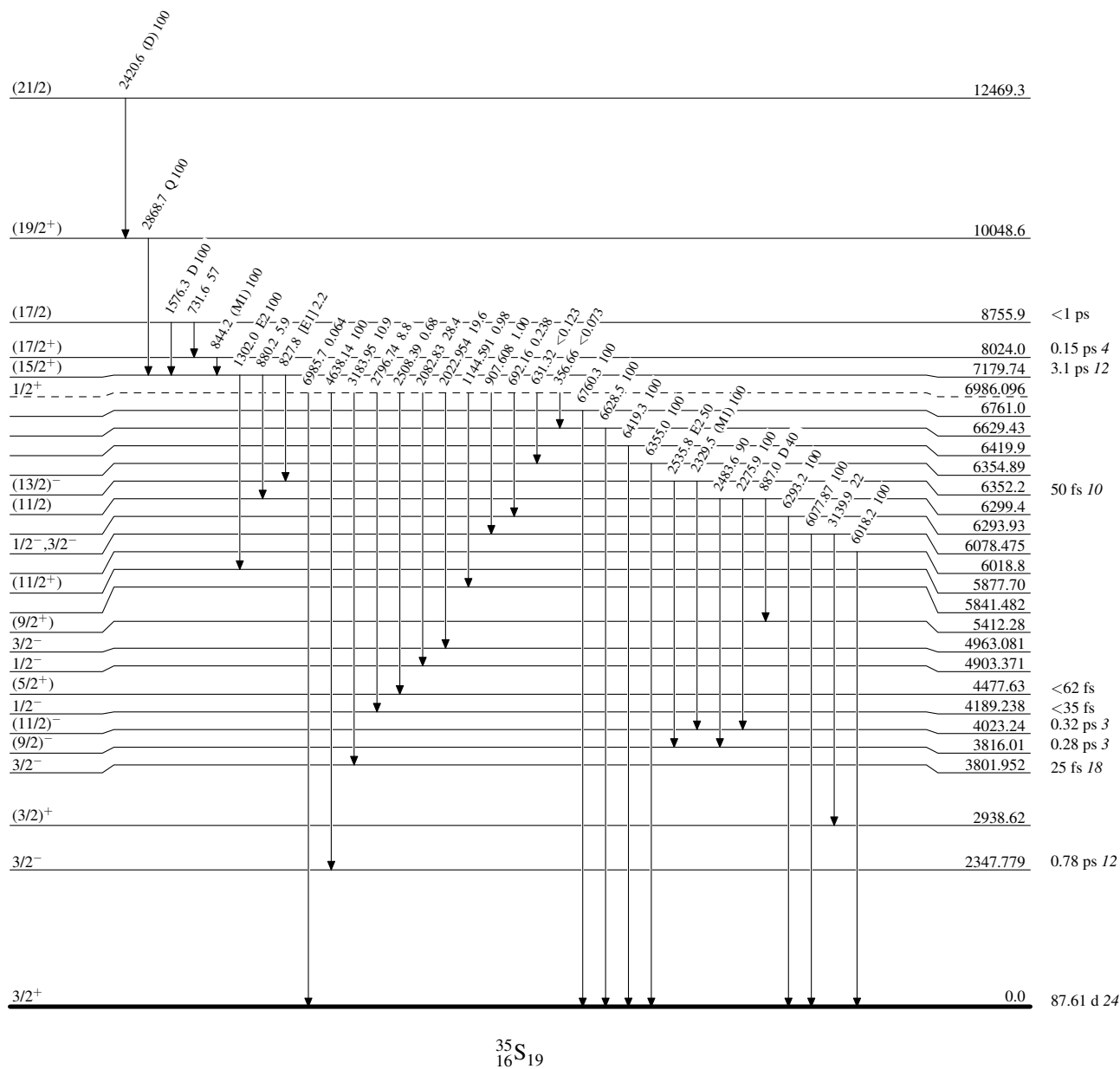
[#] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Multiply placed.

& Placement of transition in the level scheme is uncertain.

Adopted Levels, GammasLevel Scheme

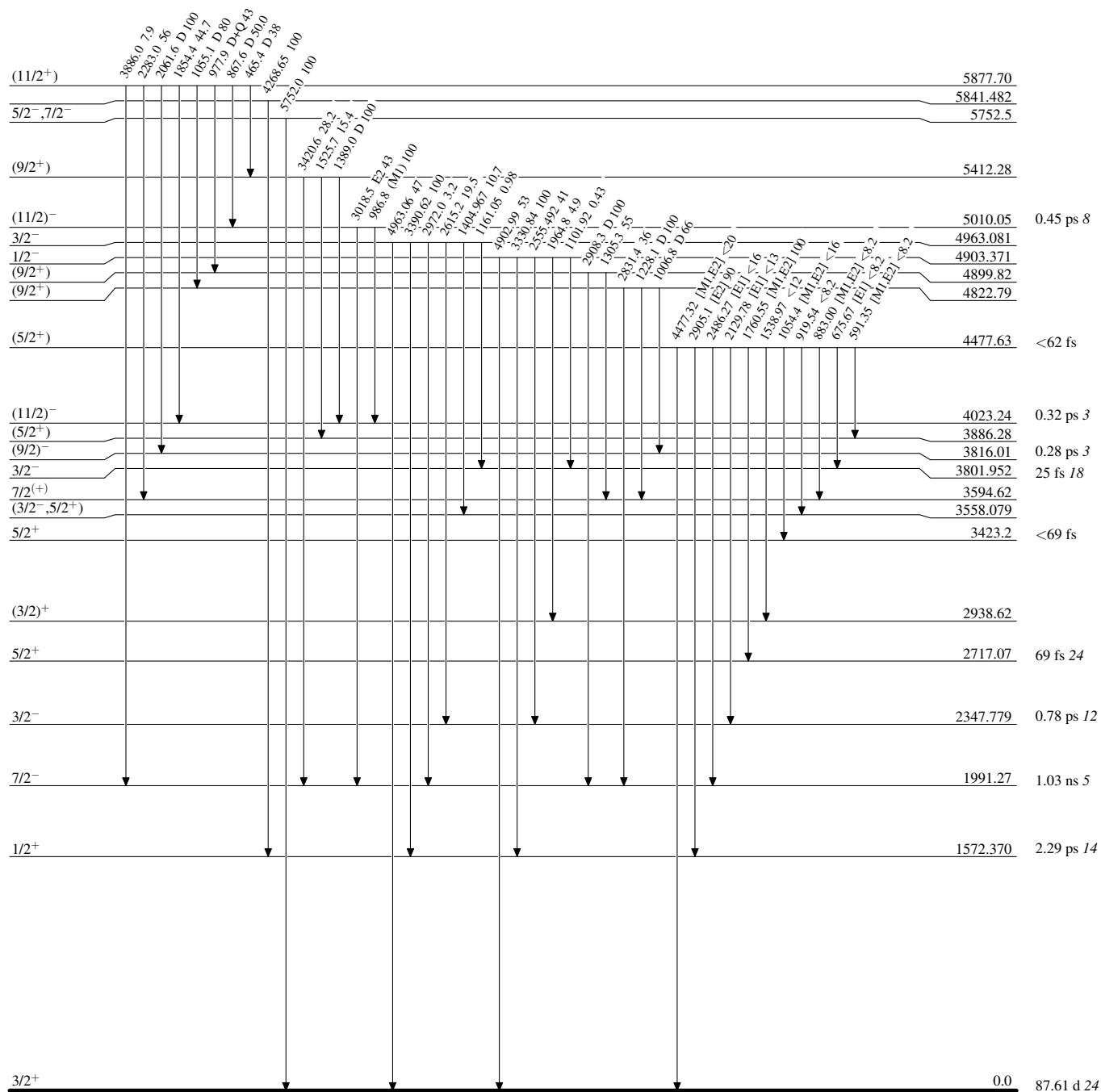
Intensities: Relative photon branching from each level

 $^{35}_{16}\text{S}_{19}$

Adopted Levels, Gammas

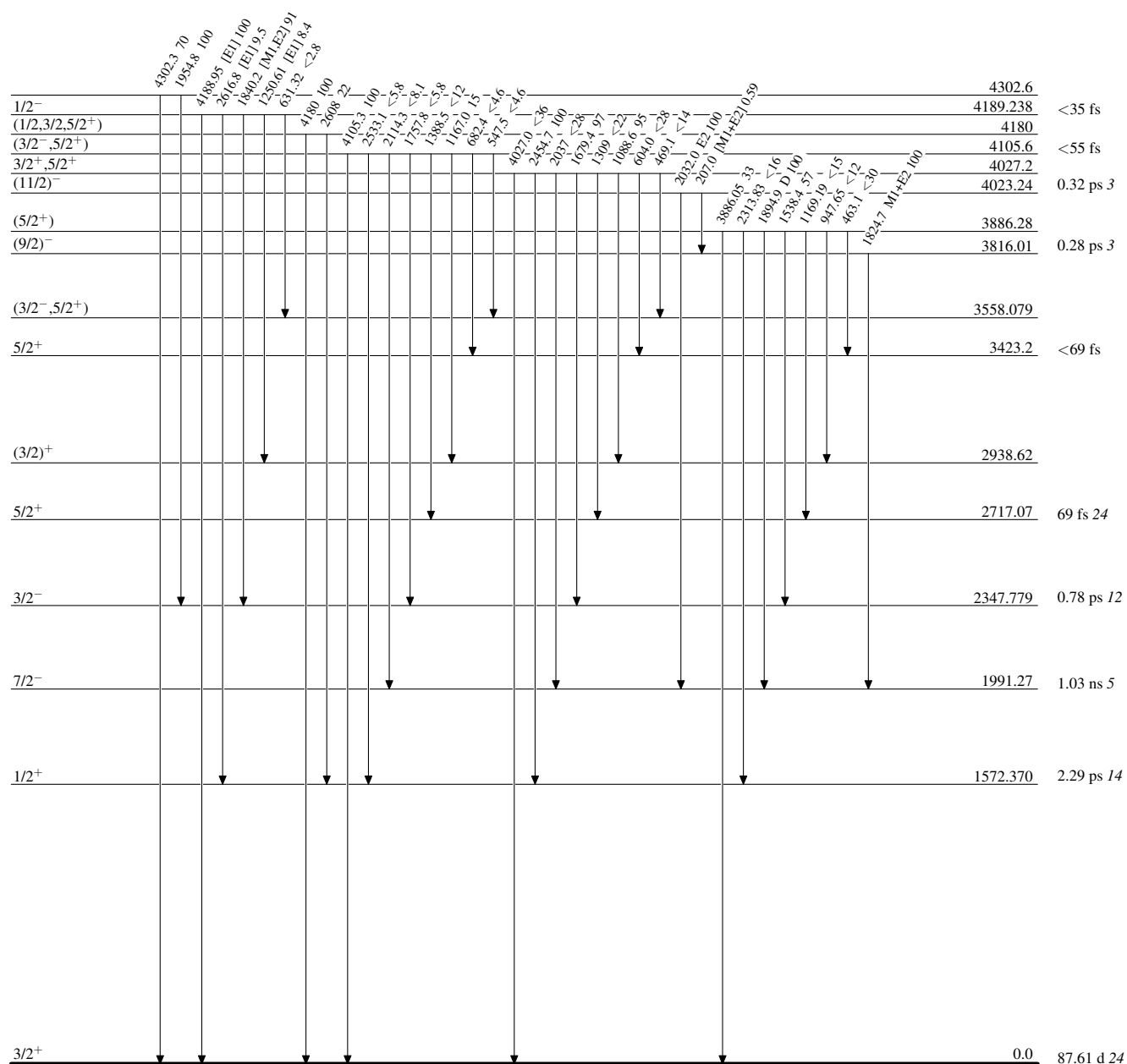
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----> γ Decay (Uncertain)

